

The SUS Experience – Genoa 2025 Edition

Stats Under the Stars 8 – YoungSIS Hackathon

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From Beach Bum to Best Project: A Statistical Miracle in Genoa

Sapienza University of Rome

Prelude: Split (with Missing Values and High Variance)

The story of SUS8 — Stats Under the Stars, 8th edition — begins before the code. Before the caffeine. Before even the pesto.

It begins with a split. Or rather, with a stratified sampling. Our team, initially one coherent cluster, was divided, not by conflict, but by logistics;

Sara, Alessia, and Giuseppina chose the optimal strategy: they left for Genoa the day before the event, minimizing travel entropy and maximizing rest. Once there, they explored the city, enjoyed spritz-powered EDA (Exploratory Drinking Analysis), and retreated to Giuseppina’s family home for a peaceful night of sleep.



Figure 1: The girls arriving early

Meanwhile, on the morning of June 15th, **Emanuele** initiated a more stochastic route. He left Rome at dawn, not with the team, but with **Roberto**—a rival-turned-ally from another group. Together, they formed a two-node graph with limited bandwidth but surprisingly high throughput.



Figure 2: The Road Trip.

Forget trains. Forget hotels. This was an **unsupervised clustering attempt** with limited constraints, one car, and unlimited storytelling. The dataset: 6 hours of highway. The features: life stories and a playlist of Napolitan Neomelodic music. As Genoa approached, the two analysts made an unexpected decision: *Let’s stop on the beach. Clear the mind before the war.*

Destination: **Boccadasse**—a place known for its colorful houses and waters so cold they’re said to reset overfitted brains.



Figure 3: Boccadasse beach, ofc it ended up raining 15 min after.

Emanuele—heroic, barefoot, slightly overfitted to optimism—slipped on the rocks and earned a bruise on his feet. A literal **penalty term** for exploring too far off-distribution.



Figure 4: Outlier detected: Emanuele vs. the scogli of Boccadasse.

But bruises heal. Like models retrain. And time, like a progress bar, keeps moving.

Act I: The Gathering and the First Refuel (a.k.a. The Lasagna Prior)

By late afternoon, Genoa was full of excitement. Our team, coming from different journeys and many stories, finally met: four distinct data points merged into a single high-performance cluster.

The hackathon venue was busy: laptops were everywhere, teams were talking, and tables were covered in snacks and chargers. Our arrival was not quiet. We were a mix of fresh energy, tired feet, ideas warmed by spritz, and a strong wish to do well.

Before the coding started, the organizers gave us an amazing meal: Lasagna al pesto—big, tasty, and perfect for a long night. There were drinks for everyone, from coffee to cocktails, keeping everyone awake and ready for action. This was more than just food; it was a tradition. The “Lasagna Prior”—a boost to help us stay strong.

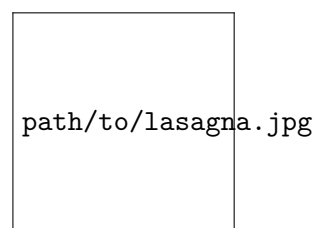


Figure 5: Lasagna al pesto: the official fuel of SUS8. (So good we forgot to do pictures)

After eating and talking, we found our seats. Every team was ready to win, every table was a challenge. But for us, the real contest was still ahead; tonight was about planning, laughing, and preparing for what was coming.

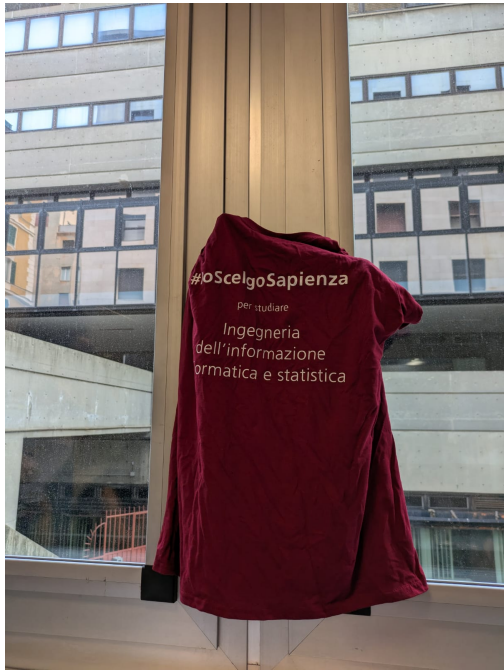


Figure 6: Questa maglia è la nostra pelle (cit. Ultras)

As the sun set over Genoa’s rooftops, everything changed. The hackathon clock was about to start. We knew soon our models and our teamwork would be tested—and so would our ability to keep going on little sleep and lots of lasagna. The stage was set, everyone was ready, and the first refuel was complete. The real competition was about to begin.

Act II: The Competition Begins

The hackathon kicked off with a clear challenge: build the best machine learning solution for money laundering detection in twelve hours. Laptops opened, snacks appeared, and the room buzzed with anticipation as the clock started ticking.

We quickly reviewed the dataset and

rules, scanning through technical papers for inspiration. With time short, we focused on strategies that were practical and effective—no room for wild experiments, just solid teamwork and quick decisions.

Sara and Alessia blazed the trail, launching our first working submission at 23:00. It ran smoothly, scored well, and, crucially, didn’t crash—a small miracle in hackathon land.

Giuseppina, steady and methodical, followed up with a model that hit an AUC of 0.70 and secured our spot on the leaderboard.



Figure 7: The moment our first submission didn’t crash: small miracles at midnight.

As the night wore on, Emanuele—caffeinated beyond safe limits—refused to stop optimizing.

"Stop coding," said the girls. *"Help us write the report,"* they pleaded. But he was in the zone.

Tuning parameters. Testing new models. Rewriting feature engineering pipelines at 3 a.m. He was no longer debugging—he was locked in a spiritual battle with scikit-learn itself.

Some changes worked like magic; others

brought chaos, bugs, and marathon reruns for microscopic gains. Coffee fueled our code. Laughter and occasional shouts kept us awake.

Each iteration brought us closer to a robust pipeline. We weren't chasing perfection—just something reliable before the deadline.



Figure 8: Focused coding and teamwork as the competition heats up.

Despite mounting exhaustion, we pressed on—motivated by hope, caffeine, and the prospect of victory. Every new submission was a step closer to our goal.

Act III: The Submission Spiral and Statistical Survival

As the hours passed, the hackathon entered its most intense phase. The leaderboard became our obsession. Scores rose and fell; each submission was a roll of the dice—sometimes a triumph, sometimes a bug-induced disaster.

We cycled through models, tweaked features, and tried every trick we knew. Some runs took ages; others finished in minutes with little change. It was a crash course in knowing when to stop chasing tiny improvements.

Emanuele, still in coding overdrive, was unstoppable. The girls balanced frantic

bug fixes with cleaning code and preparing the final report.

Every small win was celebrated; every crash brought groans—and more coffee. Sometimes we argued over approaches; sometimes we just laughed at the madness. Teamwork kept us going.



Figure 9: Leaderboard obsession: every new score meant hope or heartbreak.

Giuseppina made sure submissions were strategic and well-timed, while Sara and Alessia polished the code and report.

Fatigue set in as sunrise approached—some crashed on sofas, others powered through on espresso. The final moments were a blur: last-minute submissions, frantic bug fixes, and double-checking the report.

We handed everything in just before the deadline, amazed we'd survived the night.

Act IV: Rest, Reunion, and the Ceremony

Once the competition was over, everyone was completely exhausted. Sara, Alessia, and Giuseppina went back to Giuseppina's home, finally getting some much-needed sleep after the long night.

Meanwhile, Emanuele—still running on adrenaline—left with Roberto, his rival-turned-teammate. With no place to stay, they ended up sleeping on the beach, listening to the waves and trying to recover from the marathon coding session. It wasn't the comfiest bed, but it felt like a true adventure.



Figure 10: Emanuele and Roberto, the beach as their bed after the hackathon.

Act VIII: The Main Event — Ceremony Chaos and Victory

The awards ceremony was the scene of our final showdown. We arrived from separate worlds: the girls rested and refreshed after sleeping at home, Emanuele and Roberto a bit sandy and bleary-eyed from their night on the beach.

The room was packed, everyone buzzing as the judges prepared to announce the winners. Our team gathered near the back, half-expecting nothing, half-hoping for something.

Then, the results were revealed: Our team was called for the **Best Project a.k.a. FIRST PLACE!** There was a burst of cheers. The girls shouted and hugged while Emanuele held a Thug Life pose. We went up to the stage—half awake, half in disbelief—to collect our prize.



Figure 11: The moment of victory: best project award!

But the surprises weren't over. Out of nowhere, Emanuele was greeted by an old family friend—actually, the mother of one of his childhood friends—who works for IS-TAT and was attending the event. She snapped a victory photo and sent it to her son, making the moment even more memorable.



Figure 12: The picture in question

Now, our next goal is clear: build a new project for ISTAT (EHIS2019 we are bringing to the exam of StatLearn), hopefully impress her enough to land a job there for EHIS2025 in November!

Right after the ceremony, the organizers stopped us for an interview. We were all a bit shy, but Giuseppina confidently took the stage and explained our hackathon journey and the winning project.

We tried to stick around for some networking, but exhaustion won. You can see it in the photos—hollow eyes, tired smiles, but pure happiness. The girls returned to Giuseppina’s home for a final rest and then caught the night bus to Rome, sleeping the whole way. Emanuele and Roberto drove straight back to Rome.

Back in Rome, we celebrated with one last drink and announced, “Guess what? We just won the statistics hackathon in Genoa and brought home €1000!” It was a fun ending to an unforgettable adventure.

data should not be reused or shared outside the context of the event.

Special Thanks: A huge thank you to the SUS8 organizers, SIS, and all the sponsors for hosting an unforgettable event—and to everyone who made the hackathon fun, challenging, and inspiring. Thanks also to the mentors, judges, and fellow participants for their energy, advice, and late-night camaraderie.

For next year SUSPeople, May your next hackathon be as eventful, rewarding, and statistically significant as ours!

Appendix

All materials from our SUS8 hackathon adventure are available on GitHub:

GitHub link: <https://github.com/emanueleiacca/SUS>

In this repository you will find:

- **Data description:** Overview and structure of the dataset used in the competition.
- **Code:** All scripts, Jupyter notebooks, and pipelines for our machine learning models and analyses.
- **Report:** Documentation of our approach, experiments, and results.

Note: Please respect the privacy and rules regarding the dataset—competition